

HEARDS- Data Cleaning Pipeline

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HEARDS Dataset – Status Update 1 Overview

- Parsed 215K+ raw csv trades (text format) → structured CSV (regex & NLP extraction).
- Cleaned/standardized key fields: price_basis, volume, location, etc.
- Added & validated updatedDate (uniform format, 100% filled).
- Labeled 500 headlines with GPT → trained RF models for start_date & end_date.
- Applied model to full dataset → saved as hearsd_structured_cleaned_07.csv.
- QA: No future/invalid dates; 99%+ fill across all date columns.
- **Next:** Normalize price using historical price_basis; run advanced EDA.

Initial vs Final Dataset Comparison

- **Initial Rows:** 215,400
- **Final Rows:** 195,672
- **Retention:** 90.85% → ~9.15% dropped
 - *Excluded rows were often announcements (e.g., "Platts is monitoring...")*

Initial Dataset Columns:

- headline – trade action summary (100% filled)
- id – unique identifier (100%)
- updatedAt – publication date (100%)

Final Dataset Columns:

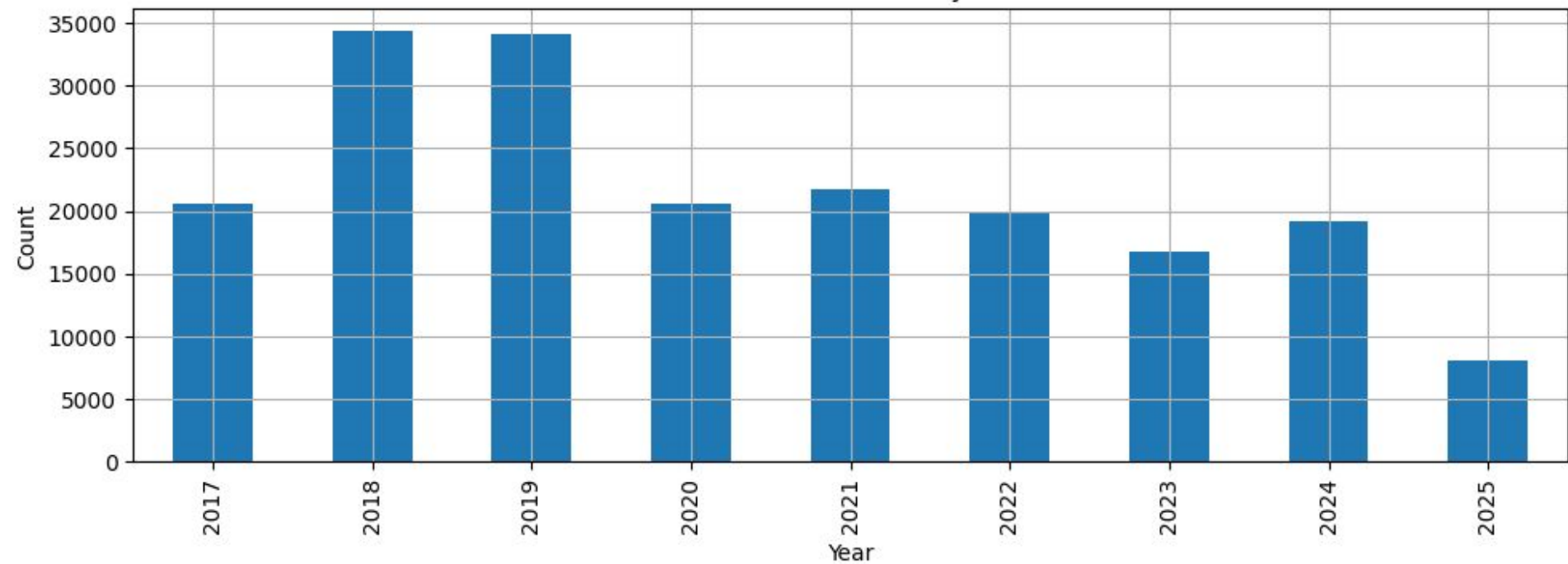
- Parsed fields added:
 - Update_date (**100%**)
 - start_date (**99.43%**)
 - end_date (**99.27%**)
 - action_type (**100%**)
 - party (**91.15%**)
 - price (**100%**)
 - price_basis (**80.92%**)
 - grade (**89.18%**)
 - incoterm (**82.20%**)
 - location (**82.74%**)
 - volume_min / volume_max (**61.09%**)
 - frequency_tag (**60.76%**)

Column Overview – Final HEARDS Dataset

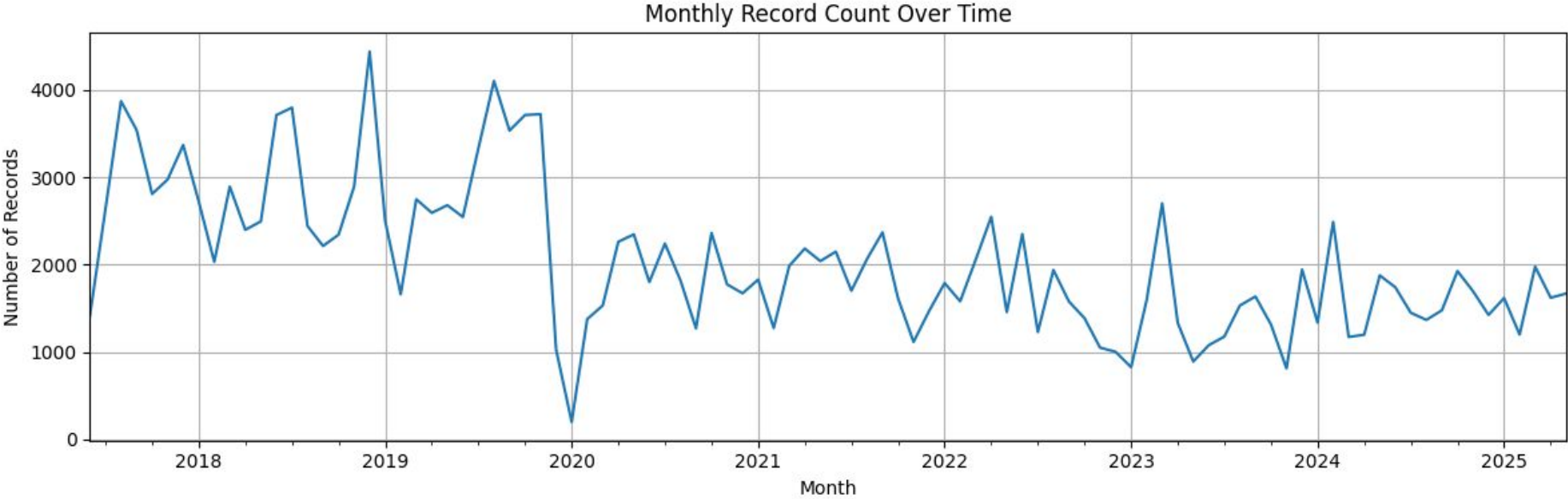
- updatedDate** – Reported date; baseline for inferring other dates.
- start_date** – Predicted start of trade window (from headline + updatedDate).
- end_date** – Predicted end of trade window (same method).
- action_type** – Nature of market move (e.g., *Offer*, *Raise*, *Lower*).
- party** – Entity initiating the action (e.g., *VITOLSG*, *GLENCORES*).
- price** – Quoted numeric value from headline.
- price_basis** – Reference price type (e.g., *MOPS 380*); for normalization.
- grade** – Commodity quality/grade (typically 180 or 380).
- incoterm** – Shipping term (e.g., *FOB*) indicating delivery conditions.
- location** – Trade location (e.g., *Straits*, *Asia*).
- volume_min** / **volume_max** – Volume range, if stated.
- frequency_tag** – Time label (e.g., *5 Day*, *BalMnth*).
- headline** – Original trade text; used in all model training and parsing.

UpdateDate

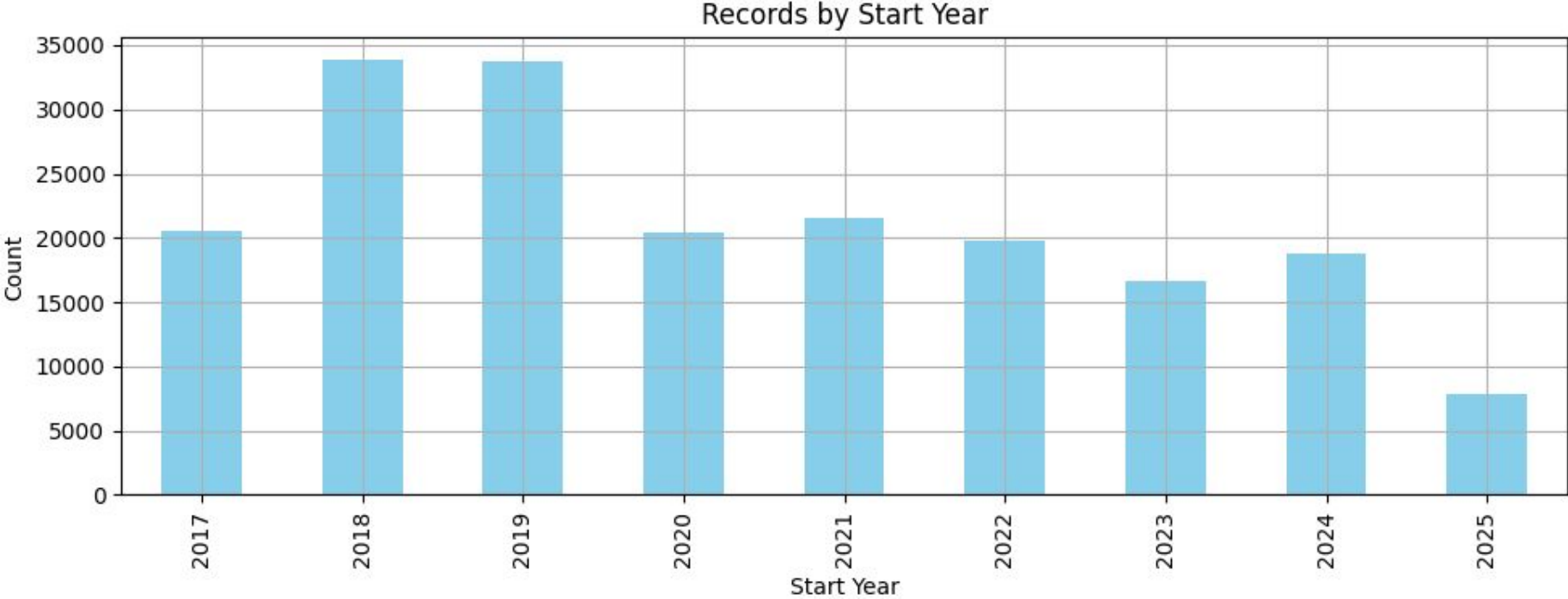
Number of Records by Year



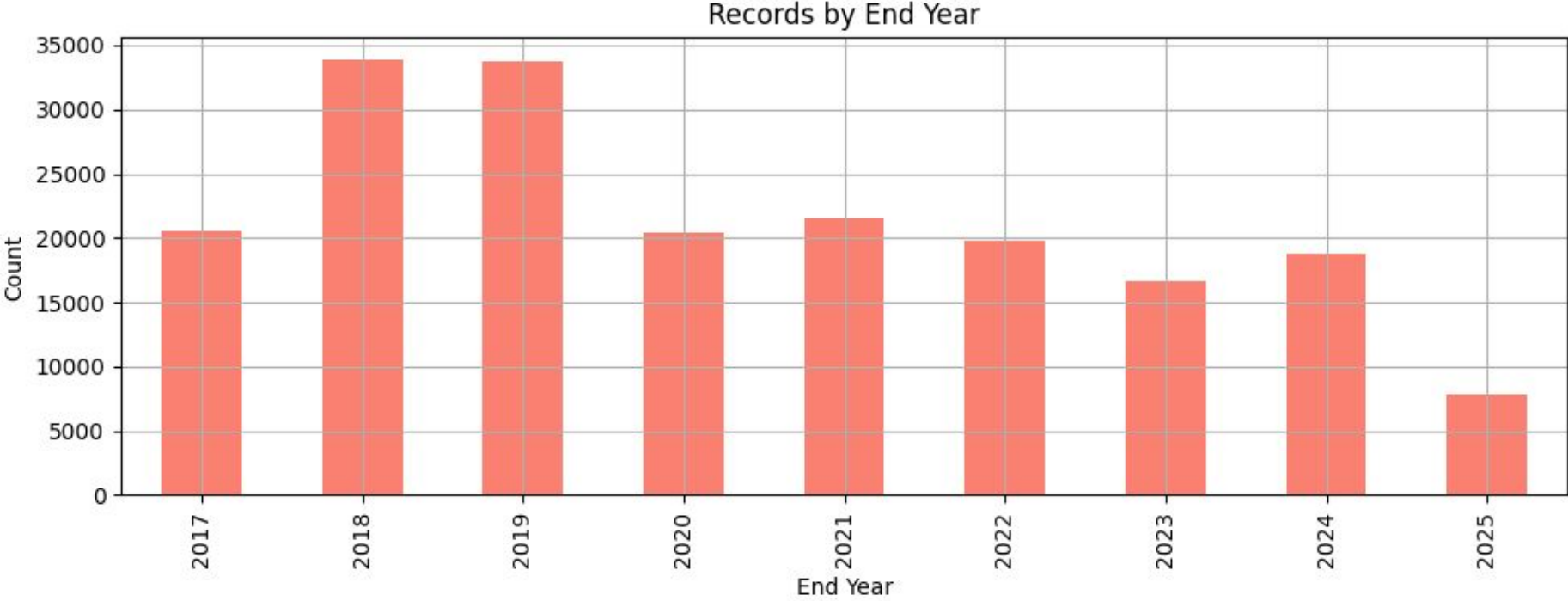
UpdateDate



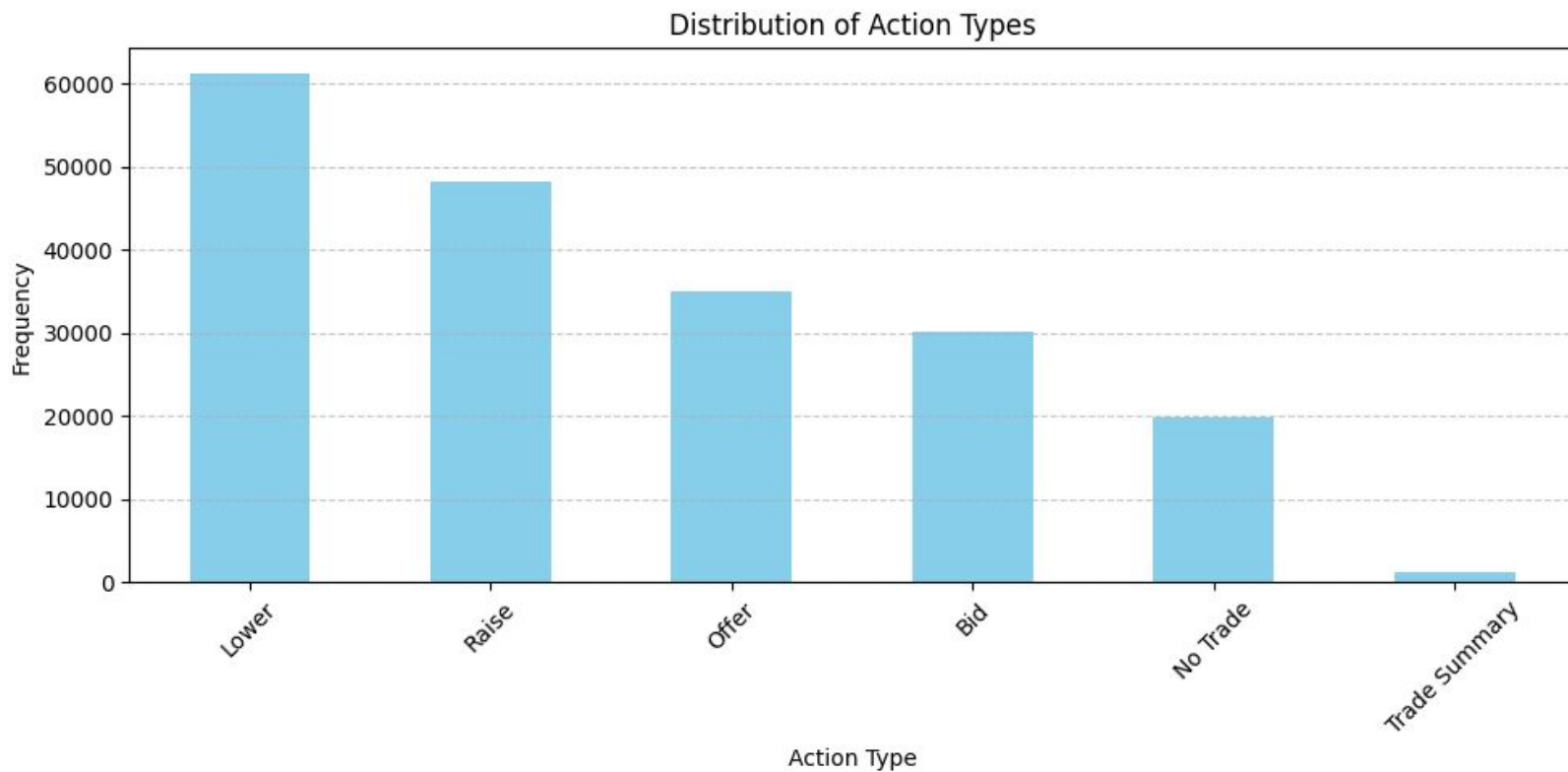
Start & End Date



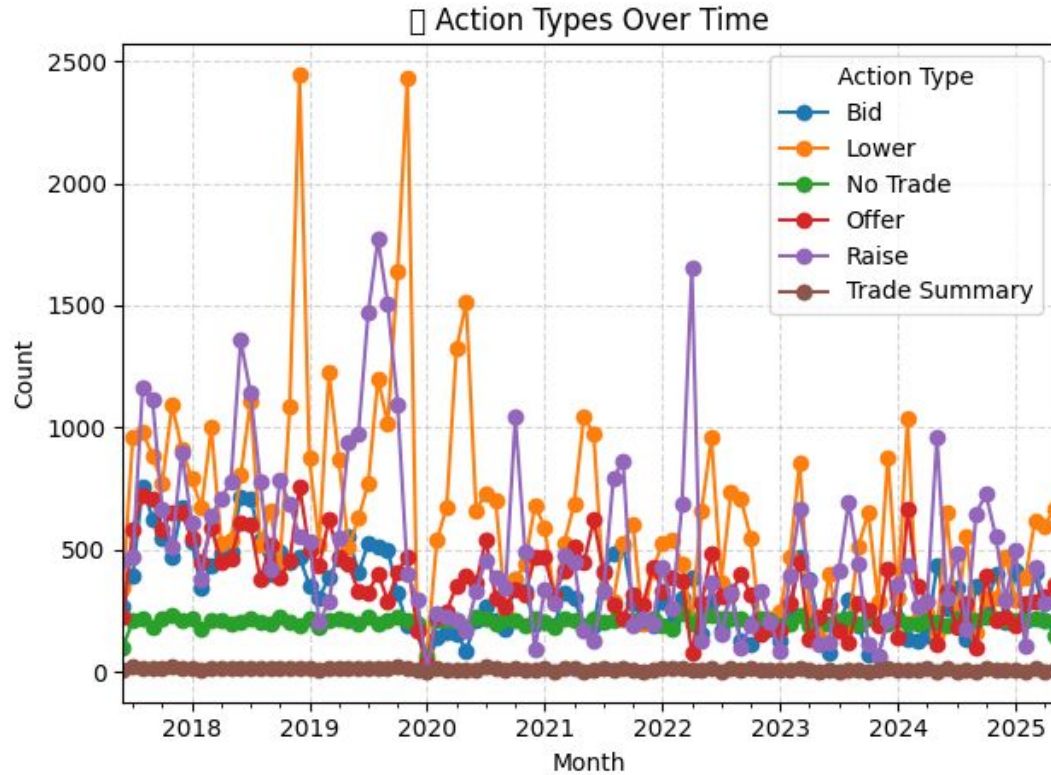
Start & End Date



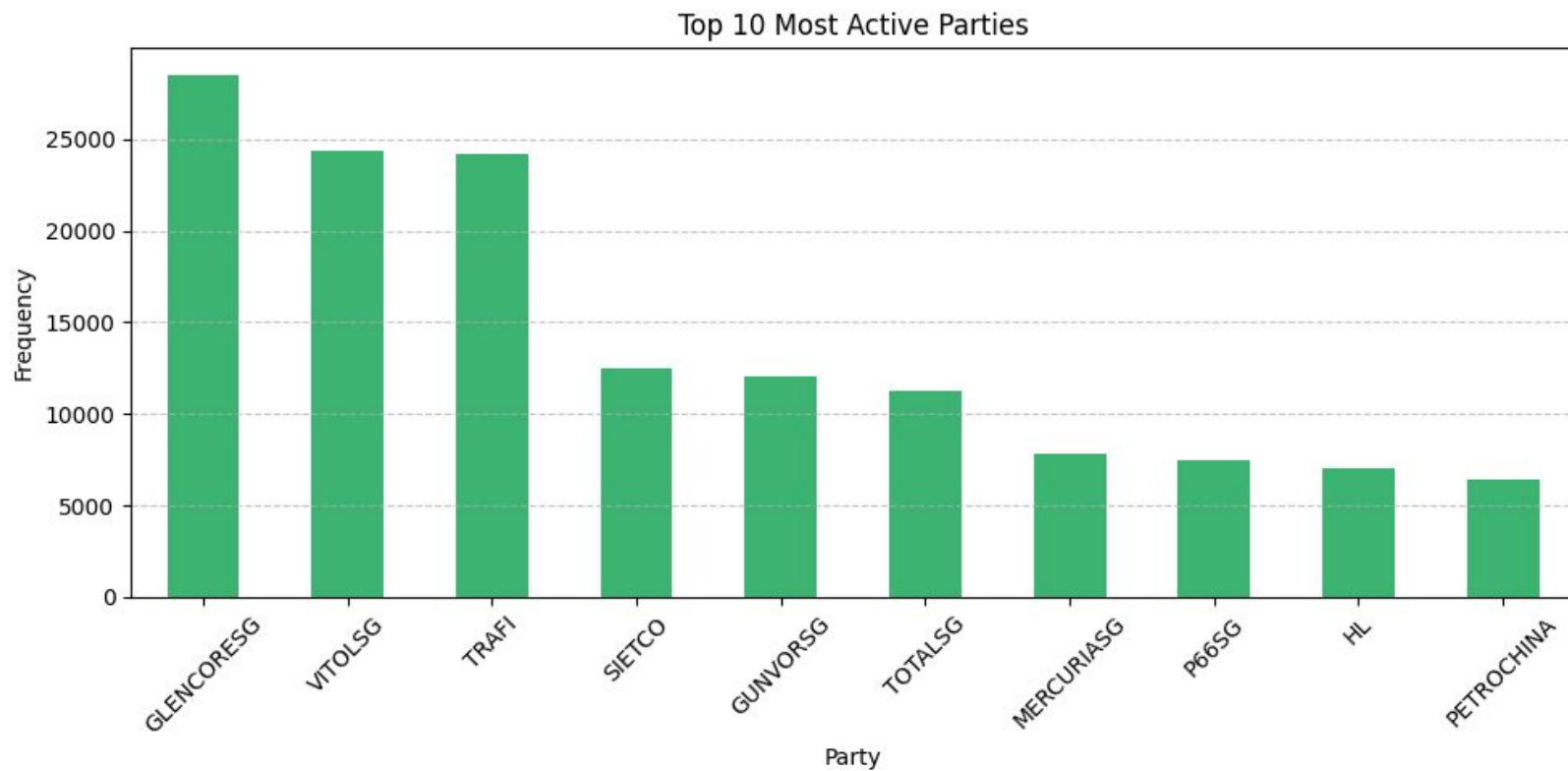
Action Types



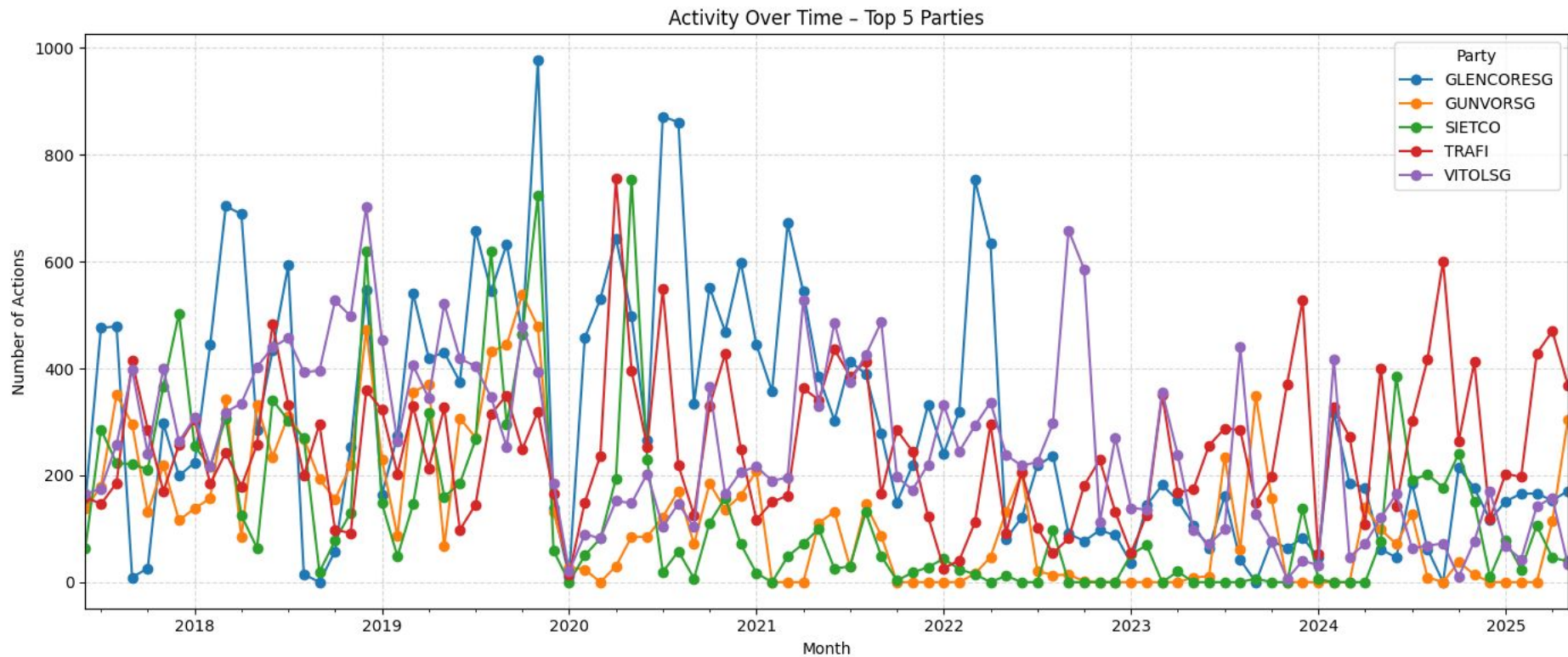
Action Types



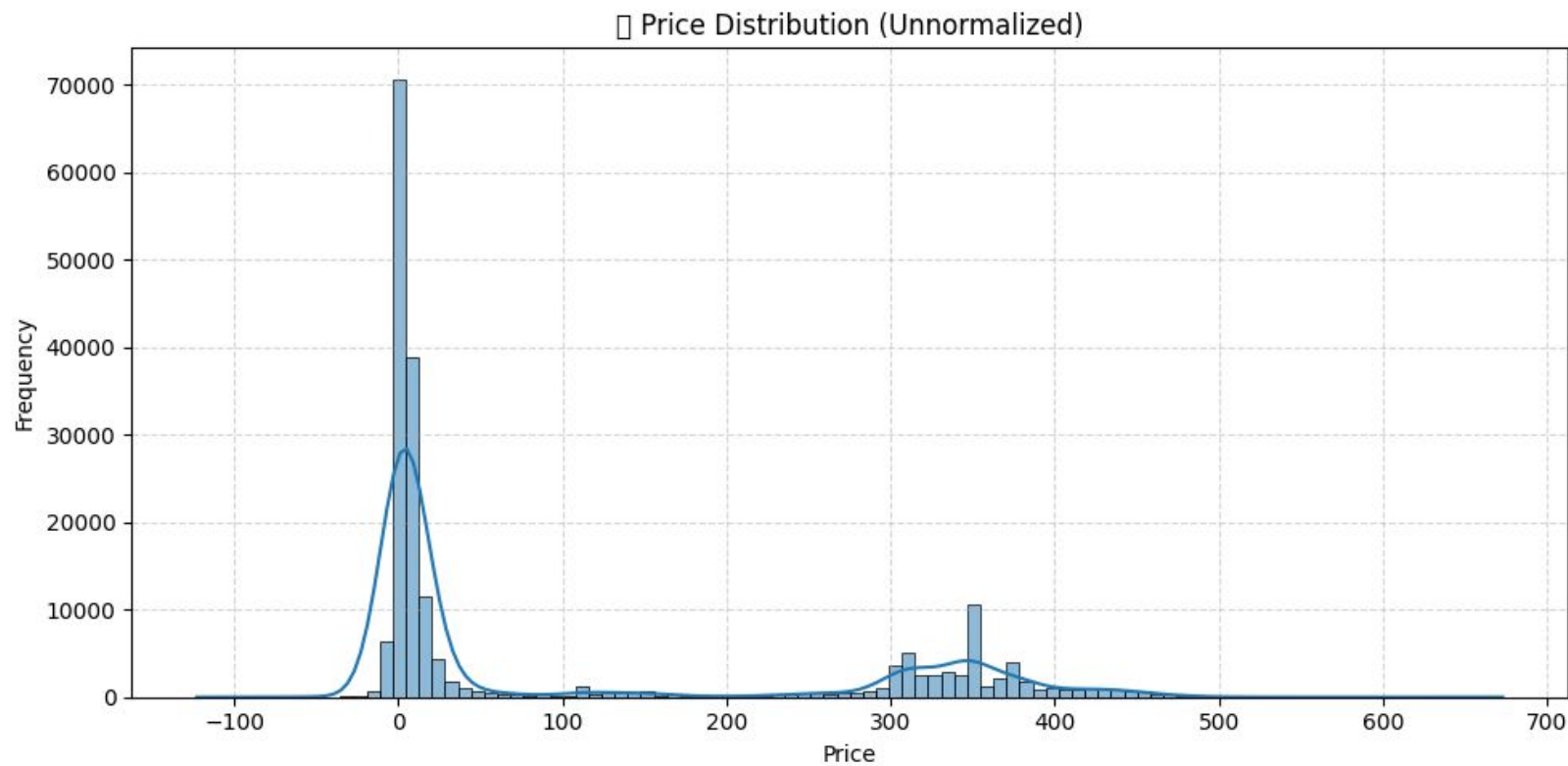
Party



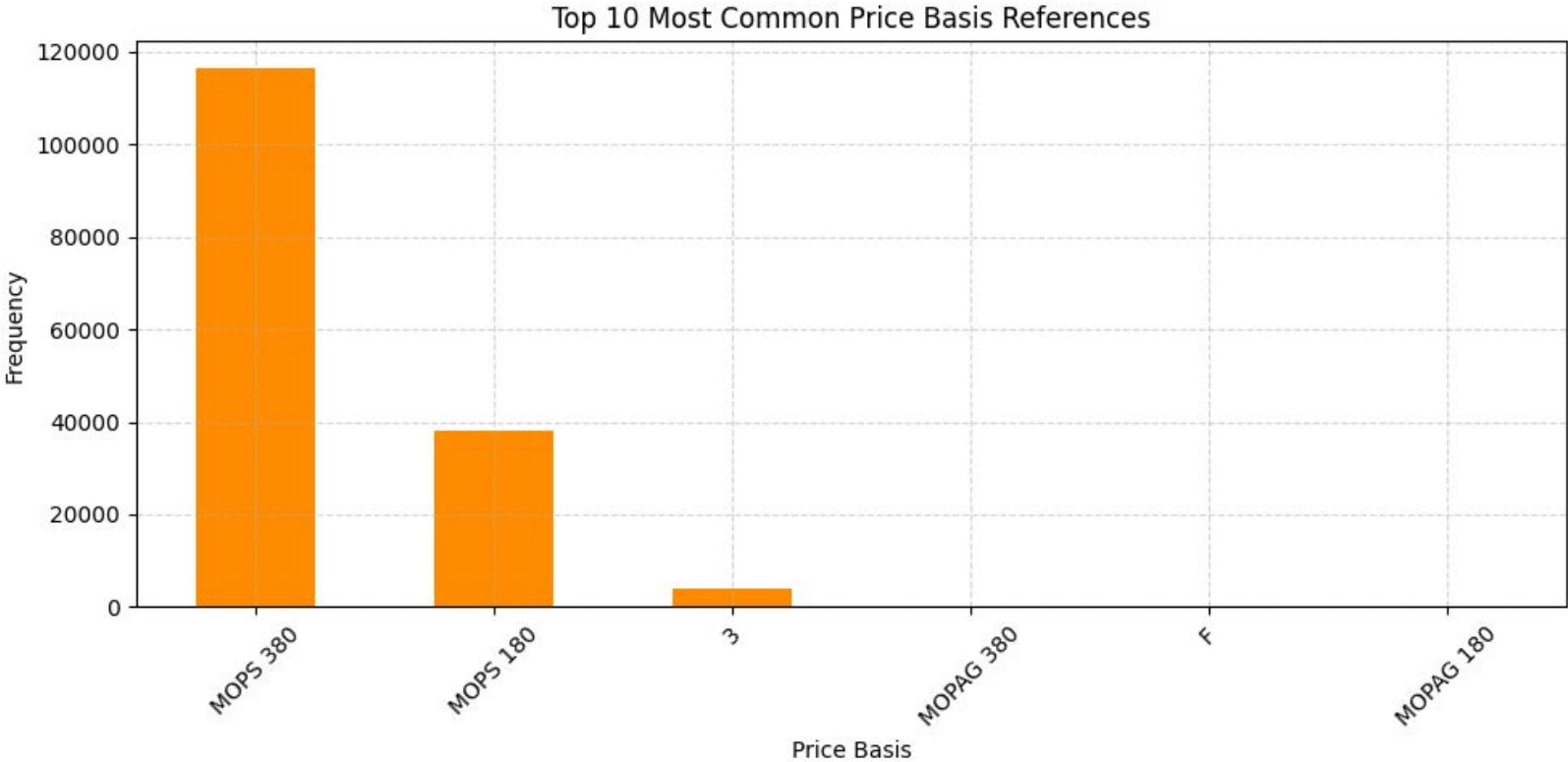
Party



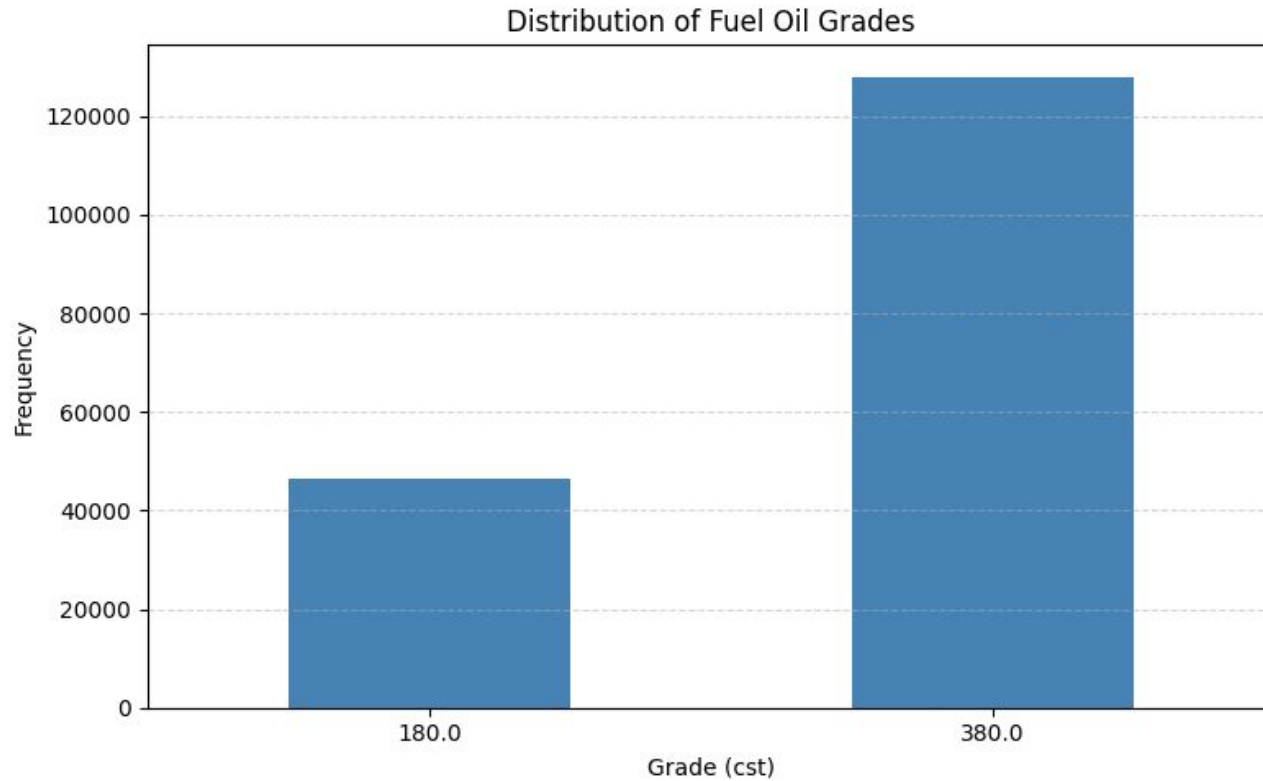
Price



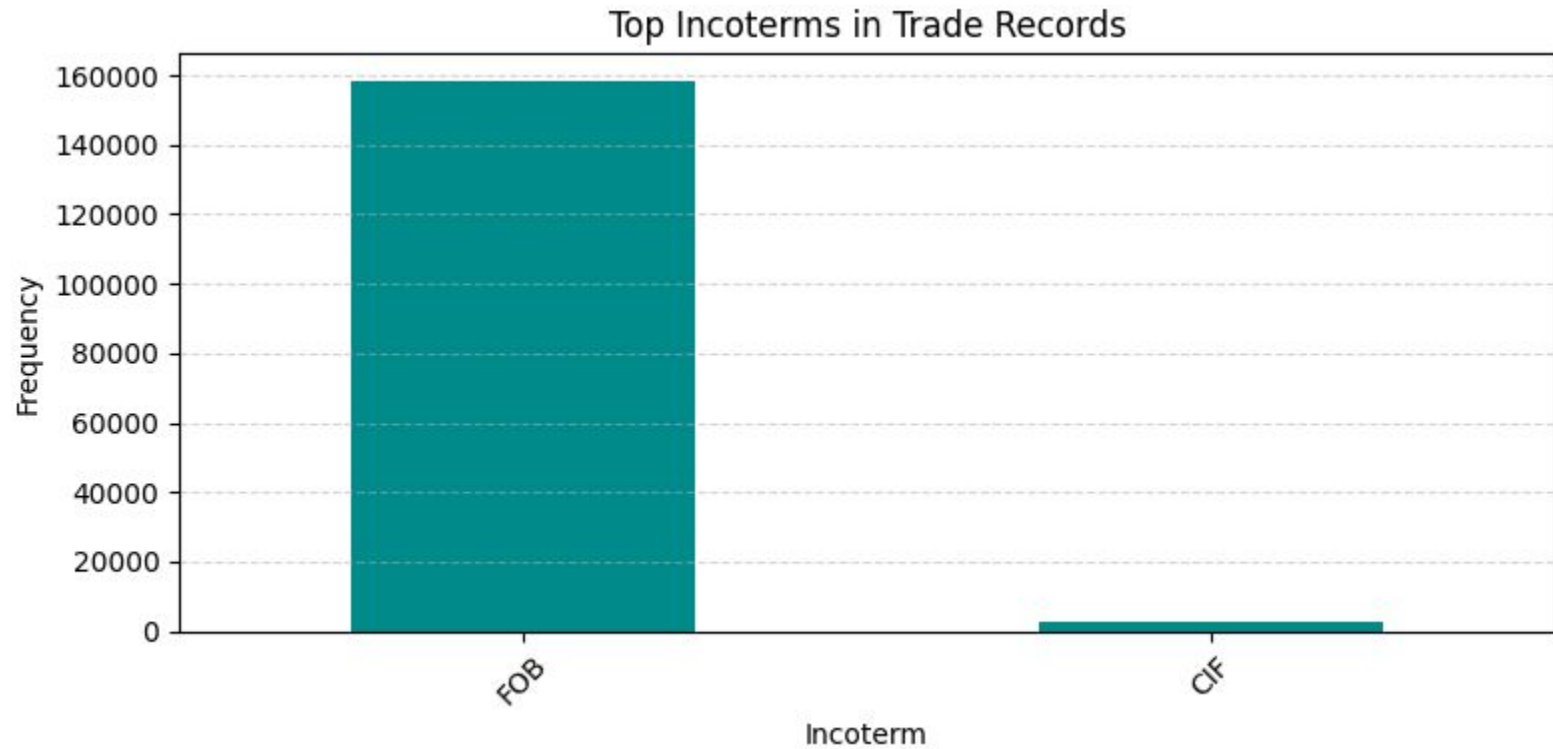
Price Basis



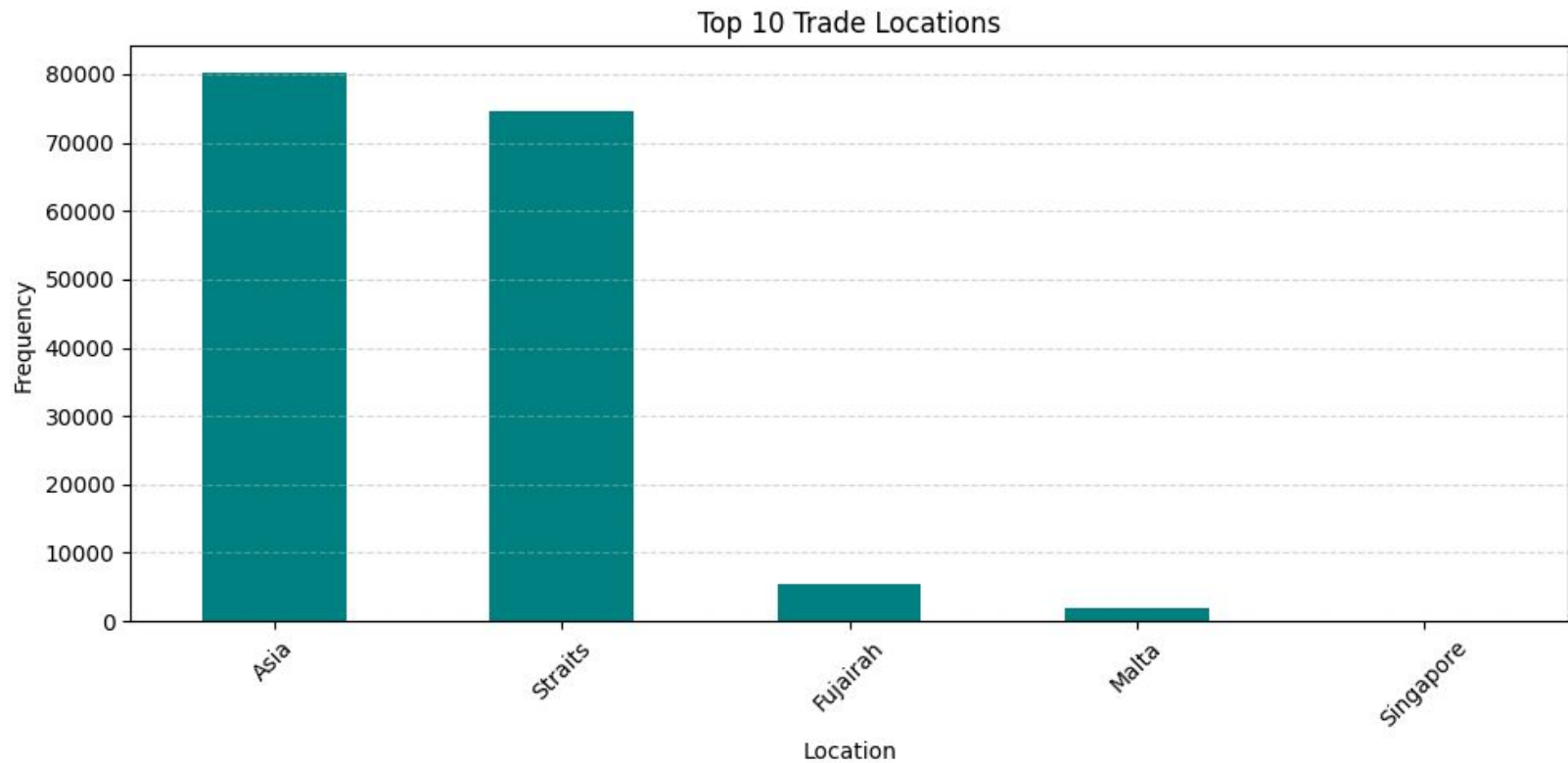
Grade



Incoterm



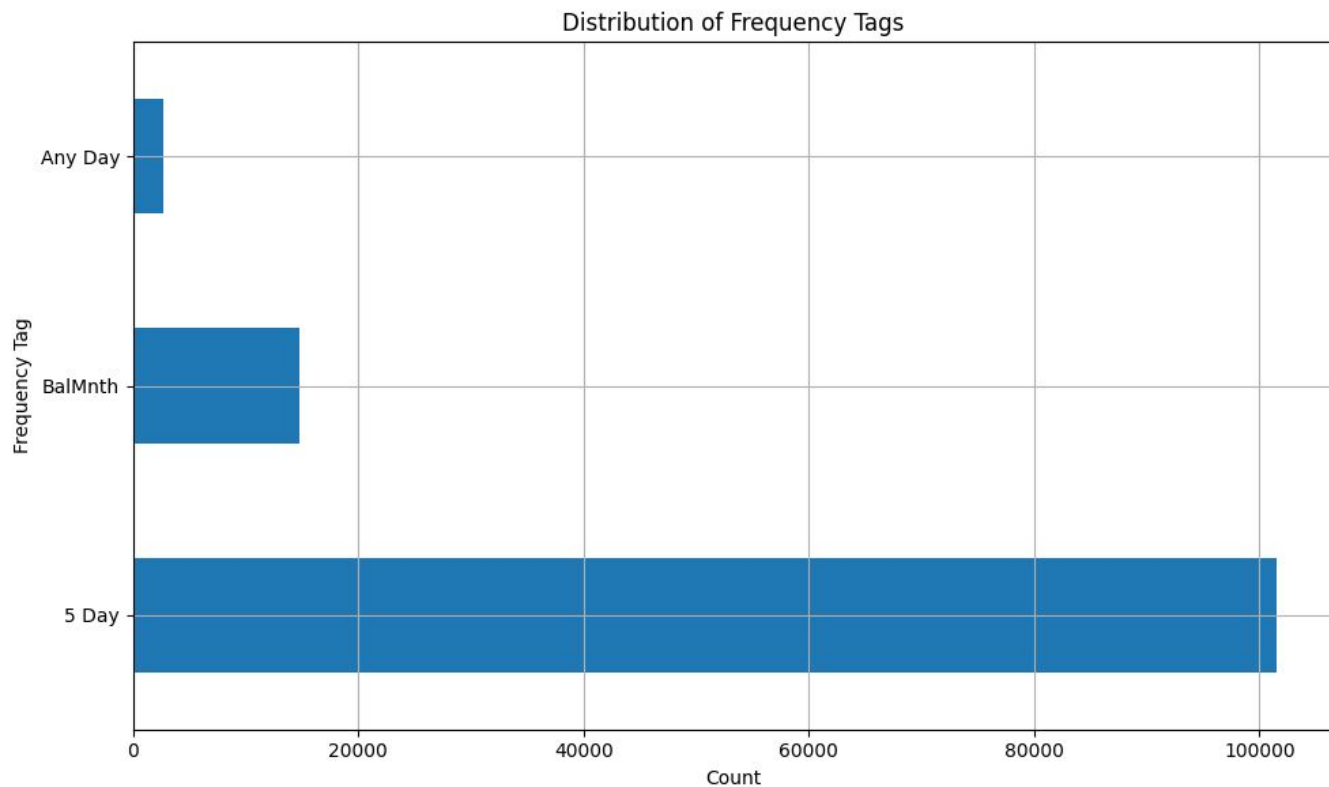
Locations



Volume

	volume_min	volume_max
count	119534.000000	119534.000000
mean	22.979596	22.316462
std	7.046852	7.948603
min	20.000000	2.000000
25%	20.000000	20.000000
50%	20.000000	20.000000
75%	20.000000	20.000000
max	40.000000	40.000000

Frequency Tag





Field	Initial Dataset	Final Dataset
headline	Platts HSFO 380cst FOB Straits 15-30, TRAFI bids Jul 18-Jul 22 100% MOPS 380 5 Day \$2.00 for 40-40 OCO 7:57:10.885 GMT	Platts HSFO 380cst FOB Straits 15-30, TRAFI bids Jul 18-Jul 22 100% MOPS 380 5 Day \$2.00 for 40-40 OCO 7:57:10.885 GMT
updatedAt	2024-07-03T07:57:10.919Z	2024-07-03
start_date	None	2024-07-18
end_date	None	2024-07-22
action_type	None	Bid
party	None	TRAFI
price	None	2.0
price_basis	None	MOPS 380
grade	None	380.0
incoterm	None	FOB
location	None	Straits
volume_min	None	40.0
volume_max	None	40.0
frequency_tag	None	5 Day